



Need for Materials and Energy

Think of what an organism needs as it grows and develops. Just as a building grows taller because workers use energy to assemble new materials, an organism uses energy and a constantly supply of materials to grow, develop, and reproduce. Organisms also need materials and energy just to stay alive. The combination of chemical reactions as it carries out its life processes is called **metabolism**.

Response to the Environment

Organisms detect and respond to stimuli from their environment. A **stimulus** is a signal to which an organism responds. External stimuli, which come from the environment outside an organism, include factors such as light and temperature. In contrast, internal stimuli come from within an organism. This field of sunflowers is responding to the stimulus of sunlight.



Maintaining Internal Balance Even though conditions in the external environment may vary widely, most organisms must keep internal conditions, such as temperature and water content, fairly constant to survive. The relatively constant set of conditions is called **homeostasis**. Often internal stimuli help maintain homeostasis. For example, when your body needs more water to maintain homeostasis, internal stimuli make you feel thirsty.

Evolution Although individual organisms experience many changes during their lives, the basic traits they inherited from their parents usually do not change. As a group, however, any given kind of organism can **evolve**, or change over time.

Over a few generations, the changes in a group may not seem significant. But over hundreds of thousands or even millions of years, the changes can be dramatic. The ability of a group of organisms to change over time is invaluable for survival in a world that is always changing.

What About Viruses? Viruses are particles made up of proteins, nucleic acids, and sometimes lipids. Viruses depend entirely upon other living organisms for their existence, making them parasites. After infecting living things, viruses can reproduce, regulate gene expression, and evolve.

 **READING CHECK Explain** Why is it important for organisms to maintain homeostasis?